

# Analysis of Web Filtering Solutions

*December 2, 2019*

## Background

To maintain compliance with the Children's Internet Protection Act of 2000, schools and libraries receiving Federal e-Rate funding must employ technology protection measures to prevent access by students to material that is obscene, child pornography, or harmful to minors. Since 2015, the school district has used a Barracuda 910 Web Security Gateway as the primary Internet filter to fulfill this requirement. The purchase of this device in the spring of 2015 included a five-year support contract, with software and filter database updates in addition to technical support and a hardware warranty. This agreement expires in the summer of 2020, providing an opportunity to revisit the district's approach to web filtering.

For the 2015-16 school year, the district began a 1:1 Chromebook program, in which students are assigned computing devices that they use both at school and at home. Initially, these devices were only filtered at school, and parents were given both the ability and the responsibility to make their own filtering decisions for the devices used in their homes. This quickly proved to be an impractical approach. Most parents did not have the knowledge or skills to effectively manage the devices in their homes, and the installation of home filtering software necessitated settings on the Chromebooks that made bypassing the school filter trivial. In 2016, the district began using the free version of the Securly web filter on the 1:1 Chromebooks. This tool allows basic filtering both at school and off campus for Chromebooks, and it has alleviated the problem of unfettered access to the Internet from school devices when students are not on campus.

This approach of using two different filters is a pragmatic one, but it has significant drawbacks. The addition of the free version of the Securly product provides some protection for devices when they are off campus, but its use complicates the overall filtering solution. There are differences between the two filter databases, meaning some sites are allowed on Chromebooks but blocked in classrooms and computer labs (or vice-versa). Reporting and logging is difficult to manage, because students may be using the different filters at different times. Additionally, troubleshooting and creating filter exceptions are a challenge because it's often difficult to determine which filter's rules are being applied in a particular instance.

With the expiration of the Barracuda support contract, it is hoped that the district can find a solution that meets these primary goals:

- Use a single filtering solution that applies to district-owned devices, both on and off campus, as well as guest devices using the school's network.

- Give better tools to parents to allow them to monitor their children's online behavior when they are not at school, and optionally give them some control over filter settings applied to their children when using school-provided devices outside of school.
- Give school officials better access to reports and logs of student online behavior.
- Improve notification services that can alert school personnel when students engage in risky online behavior (self harm, weapons, etc).

## Methodology and Analysis

Amplified IT, a vendor-neutral technology consulting company focused on K-12 education, provides a Web Filtering Checklist (<https://www.amplifiedit.com/web-filtering-checklist/>) that details 87 characteristics of web filters that schools should consider when selecting a web filter. This list was used as a starting point for the evaluation. An additional nine items were added to this list for a total of 96. These characteristics were each classified as "high priority", "medium priority", or "low priority". The 29 high-priority characteristics were used for the first round of evaluation.

Twelve filtering products were identified for evaluation. This list was generated from a number of sources, including vendors that had reached out to the district, recommendations from other schools, and postings on state and international school technology forums. Based on the technical specifications and descriptions on the products' web sites, each of the twelve products was evaluated on the 29 identified criteria. Several products were eliminated from consideration because they lacked critical functionality as specified in the primary goals. For example, the Untangle, CleanBrowsing, and nxFilter products do not support the filtering of devices when they are off campus, so they were disqualified from further consideration. Based on this initial evaluation, the product field was narrowed to four finalists: Securly, GoGuardian, iBoss, and LightSpeed Relay.

The four companies were contacted to set up webinar demonstrations of their products. Each company was provided with a list of questions ahead of time that the district would like to be addressed in the webinar. These questions were derived from the primary goals as well as the AmplifiedIT list, and included:

- Can the solution provide one filter that can be used for both on-premise filtering for all devices as well as off-premise filtering for 1:1 devices?
- Can filtering be provided for guest devices on the network without installation of software, certificates, or configuration changes?
- Does the product integrate with Google Apps for provisioning, authentication, and grouping by organizational unit?
- Does the solution support different filter rules/policies for different organizational units?
- Does the solution allow different filter rules for off-campus vs on-campus on an organizational unit basis?
- Can parents have access to student browsing activity?
- Can parents customize filtering policies for their children when the student is off campus?

- Can parents implement their own filtering technology at home? Which products can be used?
- Does the product include analysis and reporting of risky behavior? How robust is the reporting and to whom is it sent?
- Do teachers have the ability to monitor students' online behavior?
- Can counselors and principals generate reports of student online behavior?
- What is the most common method used by students to circumvent filtering rules?
- How is SSL filtering implemented?
- What are the privacy and security measures in place to prevent abuse of root certificates?
- What student data is collected, and how is it used?
- Does the product use an on-premise hardware solution?
- What is the pricing structure? How much will the product cost, both initially and on an ongoing basis?

The web conferences were conducted with each of the companies between October 30 and November 6, 2019. The companies gave overviews of their products and addressed the questions that had been posed.

All of the products have multiple tiers available. In general, the base package includes web filtering, and in most cases it also includes the analysis and reporting of online risky behavior by students (GoGuardian is the notable exception; that product now treats risk analysis as an add-on product). The tools that allow teachers to monitor and manage student online behavior, and other device management tools are typically considered add-on products. Each company provided general pricing, based on a student count of 3,700 and a device count of 5,000. The prices for all of the products are annual subscriptions, with no one-time costs.

A comparison of the products is listed in the table below, along with an assessment of the current product for comparative purposes:

| Characteristic                                                                  | Importance       | Current     | Securly          | GoGuardian  | iBoss            | Lightspeed  |
|---------------------------------------------------------------------------------|------------------|-------------|------------------|-------------|------------------|-------------|
| Simplicity of solution for filtering all needed devices                         | High             | Fair        | Good             | Good        | Good             | Good        |
| Simplicity of filtering for guest devices                                       | Very High        | Excellent   | Good             | Good        | Good             | Good        |
| Integrate with Google Apps for syncing users and policies                       | Very High        | Fair        | Excellent        | Excellent   | Excellent        | Excellent   |
| Support different rules for different groups of students                        | Critical         | Fair        | Excellent        | Excellent   | Excellent        | Excellent   |
| Support different rules for on-campus vs off-campus                             | Medium           | Good        | Good             | Excellent   | Fair             | Excellent   |
| <b>Parent access to student use reports</b>                                     | <b>Very High</b> | <b>Poor</b> | <b>Excellent</b> | <b>Fair</b> | <b>Poor</b>      | <b>Fair</b> |
| <b>Parent control of off-campus filter rules</b>                                | <b>High</b>      | <b>Poor</b> | <b>Excellent</b> | <b>Poor</b> | <b>Poor</b>      | <b>Poor</b> |
| Parents can use their own filtering products                                    | Medium           | Poor        | Poor             | Poor        | Poor             | Poor        |
| Analysis and reporting of risky behavior                                        | Very High        | Fair        | Good             | Fair        | Poor             | Good        |
| Teacher control and monitoring of student devices                               | High             | Poor        | Good             | Fair        | Fair             | Good        |
| Principal and counselor access to student use reports                           | Very High        | Poor        | Good             | Good        | Good             | Good        |
| Student data privacy policy                                                     | Very High        | Excellent   | Excellent        | Excellent   | Fair             | Excellent   |
| <b>Base cost for filtering + risk</b>                                           | <b>Very High</b> | <b>Good</b> | <b>Fair</b>      | <b>Poor</b> | <b>Excellent</b> | <b>Poor</b> |
| Cost for total solution including teacher tools and management                  | High             | Poor        | Fair             | Poor        | Fair             | Good        |
| Overall Assessment                                                              |                  | Poor        | Good             | Fair        | Fair             | Fair        |
| Base cost per student per year (filtering + risk)                               |                  | \$3.24      | \$4.01           | \$6.00      | \$1.35           | \$6.76      |
| Total cost with add-ons per student per year (teacher tools, device management) |                  |             | \$7.57           | \$12.00     | \$7.75           | \$6.76      |

In many ways, the products are very similar. The three highlighted characteristics are significant differentiators because they have high priorities and the products differ significantly in that characteristic.

To determine the overall assessment for each product, the analysis of each characteristic was weighted by its importance. These were totalled and compared to a perfect score (“excellent” in all categories) to find a percentage score. These scores were assigned assessment values based on quintile.

## Recommendation

Based on this analysis, the best fit for the school district is the Securly product. The only categories in which it is inferior to the other products are in its support for different off-campus rules for different students, and the cost. The primary limitation on the off-campus rules is that all devices must have the same “take-home” policy. That is, the same filtering rules apply to all devices when they are off campus. This limitation is offset, however, by the parent control feature. Parents can set their own limitations for their students that apply to the device when it is not being used at school. By giving the parents more oversight and control of their students’ behavior, it allows them to take a more active role in their children’s online activity.

Additionally, because the limited free version of Securly is currently used to filter Chromebooks when they are off-campus, the transition to the full product should be very smooth. The Chromebooks are already configured to use it, and adjustments would only need to be made for on-campus filtering with the full product.

Looking at cost, iBoss has the least expensive base cost. It’s clear, though, that iBoss also has the lowest scores of the four finalists in six of the fourteen categories. It is a much less capable solution when compared to its competitors, and its adoption would not adequately address two of the four primary goals. While Lightspeed and GoGuardian are both reasonably good solutions, their feature sets do not align with the district’s needs as well as Securly does, and the pricing for the base packages is significantly higher.

Each of the products also offers a number of add-on packages. While these add-ons differ significantly across the products, the primary functions are the ability for teachers to view and control what’s on the students’ screens, as well as some device management tools to simplify the deployment and control of 1:1 devices. In the case of GoGuardian, the ability to analyze and report risky behavior by students is also an add-on product. While these tools may be useful, they do not address the primary goals for the filtering solution.

Specifically considering Securly, there are three add-on tools. *Securly Auditor+* includes enhanced tools to detect cyberbullying, self-harm, and violence in the Google Suite beyond the detection of risky behavior in the student’s general Internet use. *Securly Chrometools* includes the teacher classroom management products including the ability for teachers to view what students are doing

online in real time, and to control the students' computers. *Securly MDM* is a mobile device manager to assist in managing Google and Apple devices. While these tools all have attractive features that could be useful to the district, they would also nearly double the cost of filtering, and would require a significant level of policy review and professional development to provide value to the district.

For the current project, it is recommended that the district implement the *Securly Filter* product without the add-on tools. This will provide the needed functionality of the full Securly product that will allow it to replace the Barracuda solution in an economical way, and it will address all of the identified primary goals. In the future, it may be worthwhile to consider the additional tools that Securly offers as add-ons if the need for such tools becomes a higher priority.